MIL-I-15676K(SH)
20 December 1982
SUPERSEDING
MIL-I-15676J(SHIPS)
21 June 1968
(See 6.6)

MILITARY SPECIFICATION

INDICATORS, SHIP'S COURSE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ship's course indicators, 60 or 400 hertz (Hz) used to indicate the master gyrocompass headings in various locations throughout Naval ships.

1.2 Classification. The indicators shall be of the following types, as specified (see 6.2.1):

	1 or 36 "Input speed" (IX or 36X)
Type A - 1X	400 Hz, single dial, bulkhead mounted with 6-inch dial.
Type B - 1X and 36X	400 Hz, single dial, bearing.
Type E - 1X and 36X	400 Hz, double dial, concentric steering.
Type F - 1A and 36X	400 Hz, double dial, concentric steering (console mounted).
Type G - 1X	60 Hz, double dial, course to-steer.
Type H - 1X	60 Hz, double dial, course to-steer (console mounted).
Type K - Miniature	60 Hz, single dial, bulkhead mounted.
Type L - Miniature IX	60 or 400 Hz, as specified (see 6.2.1), single dial, universally mounted.
Type M - 1X	400 Hz, single dial, bulkhead mounted with 11-inch dial.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6320

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Type N - 1X	60 or 400 Hz, as specified (see 6.2.1), single dial bearing, portable.
Type O - 1X	60 Hz, single dial, bulkhead mounted with 6-inch dial.
Type P - 1X and 36X	60 Hz, single dial, bearing.
Type R - 1X and 36X	60 Hz, double dial, concentric steering.
Type S - 1X and 36X	60 Hz, double dial, concentric steering (console mounted).
Type T - 1X	60 Hz, single dial, bulkhead mounted with 11-inch dial.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-915	- Cable and Cord Electrical, For Shipboard Use, General Specification for.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-I-17214	- Indicator, Permeability; Low-Mu (Go-No-Go).
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-S-22432	- Servo Motors, General Specification.
MIL-C-26482	- Connectors, Electrical, (Circular, Minature, Quick Disconnect, Environment Resisting), Receptacle and Plugs.
MIL-C-26482/4	- Connector, Plug, Electrical, Straight, Bayonet Coupling, Crimp Type, Corrosion Proof, Class J.

STANDARDS

MILITARY

MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-463	- Definitions and System of Units, Electromagnetic Interference and Electromagnetic Compatibility Technology.
MIL-STD-710	- Synchros, 60 and 400 Hertz.

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MILITARY (Continued)

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

815-1197068 - Gyro Compass Equipment, Ship's Course Indicator, Gimbal Ring and Bracket.

9000-S6504-73687 - Dial Markings for IC, Telegraph and Indicating Systems.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the document which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The ship's course indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 General requirements. Any type of indicator shall require external power of only one frequency, either 60 or 400 Hz, 115 volt (V), in accordance with 1.2. In addition to the requirements specified herein, the indicators shall be in accordance with MIL-E-16400.

3.3 Design. Ship's course indicators shall consist of an enclosure, housing dials, synchro mechanisms to drive the dials, and an illumination circuit. The enclosure for all types of indicators, except for types B and P, shall be either of metal or plastic. The enclosure for types B and P indicators shall be of metal. The size of the indicator shall be held to a minimum in keeping with the dial size and bezel ring dimensions. The synchro

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mechanism shall be easily removable from the enclosure without the use of special tools. For electrical connections between the synchro mechanism and the enclosure wiring, a plug or connector attached to the synchro assembly by a flexible cable, which can double as a test cable, shall be used. A grounding conductor shall be included in the cable. Grounding shall conform to the requirements for electronic equipment in accordance with MIL-STD-1310. All internal assemblies shall be removable from the enclosure at the dial end. For indicators having plastic enclosures, a grounding strap shall be afforded in proximity to one of the external mounting holes for easy attachment to the mounting bolt used. The weight shall not exceed the following:

	Pounds
Type A and O	35
Type B and P	35
Type E and R	35
Type F and S	30
Type G	35
Type H	30
Type K	15
Type L	15
Type M and T	40
Type N	15

3.4 Synchro mechanism. Dials shall be positioned by a synchro mechanism consisting of standard Navy synchro control transformers (size 18CT6 or 23CT6 for types G, H, O, P, R, S and T; size 18CT4 for types A, B, E, F and M), an all-transistor servo amplifier, and a servo motor, except where otherwise specified herein. The mechanism of similar types of indicators covered by this specification shall be electrically and mechanically interchangeable. Synchro capacitors shall be required in accordance with MIL-E-16400. Synchros for all types of indicators shall be in accordance with MIL-STD-710. The change of servo motor shall be approved by NAVSEA if the motor is different from the one used for qualification inspection. Servo motors shall conform to MIL-S-22432 when operating in a high ambient temperature of 100 degrees Celsius (°C). Allowable temperature rise of the servo motor when operating at the high ambient temperature of 100°C shall be 65°C. Stall torque of the servo motor at the high ambient temperature shall be twice the maximum torque required for the indicator, to meet its temperature and humidity accuracy test.

3.4.1 Parts and units in the equipment shall be protected against transients from all sources. The profile of the transients to be protected against shall include a maximum voltage peak of 700 V and having a duration of 2 microseconds.

3.5 Accuracy. The allowable error for the various types of indicators shall be as follows:

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Maximum allowable error in degrees, measured in accordance with 4.5.1 and 4.5.2, and observed on the dials

Type	Static	Dynamic	
	One speed dial	Six speed dial	One speed dial
A and O	0.5	---	1.0
B and P	0.2	---	0.5
E and R	0.5	0.15	1.0
F and S	0.5	0.15	1.0
G	0.75	---	1.5
	(Inner dial) 0.50	---	1.0
	(Outer dial)		
H	0.75	---	1.5
	(Inner dial) 0.50	---	1.0
	(Outer dial)		
K	1.0	---	2.0
L	1.0	---	2.0
M and T	0.5	---	1.0
N	1.0	---	2.0

3.5.1 The accuracies specified herein for synchro motor driven dials is for operation with synchro transmitter of identical size.

3.6 Degree of enclosure. Except as otherwise specified herein, the enclosures for the ship's course indicators shall be watertight (3-foot).

3.7 Illumination. Dial illumination for all types of indicators shall be red internal illumination in accordance with MIL-E-16400 except as specified in 3.12.4.4 and 3.12.6.4.

3.8 Synchronization. The indicators shall be self-synchronous from all positions out of synchronism with the input signal (including the 180-degree position, except for the 1-speed indicators).

3.9 Oscillations. The indicators shall follow the motion of the input transmitters smoothly and with no spurious oscillations.

3.10 Permeability. Parts except those necessary to electromagnetic circuits, ball bearings, shafts, thrust plates, and similar application where strength or hardness is essential, shall be constructed of nonferrous, non-magnetic material having a permeability no greater than 2.0.

3.11 Painting. The interior and exterior surfaces of the ship's course shall be painted in accordance with MIL-E-16400.

3.12 Detail requirements.

3.12.1 Types A and O, 60 and 400 Hz, single dial, bulkhead mounted indicators. The types A and O indicators shall mount directly on a bulkhead with the dial in a vertical position.

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3.12.1.1 Dial. The inner dial shall be positioned by the synchro mechanism, and shall be in accordance with Drawing 9000-S6504-73687. The diameter of the inner movable dial and the effective outer diameter of its degree mark circle shall be 6.0 inches. A white fixed and red illuminated index shall be provided in lieu of a fixed outer dial against which to read the ship's true heading. The index shall have the shape of an inverted triangle with its base and height each a nominal 1/4 inch. The inner movable dial shall have a black background with white markings. The index shall be on a black background.

3.12.2 Types B and P, 60 and 400 Hz, single dial, bearing indicators. The types B and P indicators shall be so constructed that when a Navy azimuth circle or bearing circle is attached, the cross hair shall be plainly visible against the background of the dial in any position, in daylight, twilight, and darkness. From the extreme top surface of the bezel ring to the faces of the dials, the distance shall be held between 3/4 and 1-1/4 inch.

3.12.2.1 Dial. The inner dial shall be in accordance with Drawing 9000-S6504-73687 as to markings; shall have all markings appear outside the periphery of a circle 5-1/4 inches in diameter, and shall be positioned by a synchro unit consisting of two control transformers, a single channel amplifier with input selector circuit, and a servo motor arranged to provide thirty-six speed accuracy on a single self synchronizing dial. Around the moving dial shall be a fixed ring dial graduated from 0 degree to 359 degrees in one-degree divisions with markings corresponding to those for the inner dial, but with the numerals to the outside edge. This relative bearing dial shall be used in determining the bearing of an object relative to the ship. The dials shall have a white background with black graduations to facilitate the taking of bearings stroke width for minor graduations shall be .012 x 1/8 inch and 5 and 10 degree graduations shall be .015 x 5/32 inch. Provision shall be made so that by simple adjustment the indicator may be operated with just a 1-speed input in lieu of the 1 and 36-speed input.

3.12.2.2 The indicator shall have a rim of 70 percent - 30 percent copper-nickel alloy. The rim shall have a diameter of 7.800 inches +0.000/-0.005 inch, to permit use of a Navy azimuth circle or bearing circle.

3.12.2.3 The top bezel ring of the bearing indicator shall be beveled at the inner edge, and shall be marked to provide a fore and aft line, parallel to the lubber line of the repeater, for use with Navy azimuth circle or bearing circle for the taking of relative bearings. The fore and aft points of the bezel ring and case shall be engraved "F" and "A", respectively. The bezel ring horizontal bearing surface for the azimuth or bearing circles shall be approximately 0.390 inch in width.

3.12.2.4 Bearing indicator mounting. The case of the bearing indicator shall be so that it can be mounted in a gimbal ring and bracket mounting designed for installation on a pedestal. The gimbal ring and bracket mounting shall be as shown on Drawing 815-1197068. This mounting shall permit free motion of the bearing indicator during a roll of 45 degrees or pitch of 15 degrees from the vertical or any combination of the two, there shall be no obstruction in the line of sight when taking bearings under condition of 20 degree roll and 5 degree pitch. The furnishing of a pedestal, gimbal ring and bracket mounting or gimbal ring and bracket mounting only, for each bearing indicator, shall be as specified (see 6.2.1). The bearing indicator shall be

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pendulous when mounted in the gimbal ring and while having a 7-pound weight mounted on the top bezel of the indicator with the weight having a vertical center of gravity 2 inches above the top of the top bezel ring of the indicator. From the bottom surface of the pedestal mounting feet to the top of the bezel ring of a mounted type B or type P indicator, the distance shall not exceed 4 feet 6 inches.

3.12.2.5 Alinement. Dowels or alinement marks shall be used to allow disassembly of the indicator and its reassembly, without allowing more than plus or minus 0.0025 inch shift in the relative bearing dial index or lubber's line relative to the ship's reference fore and aft line bench mark. Similarly, dowels or alinement marks shall be used to prevent more than plus or minus 0.0025 inch shift between the line defined by the fore and aft marks on the bezel and the ship's reference fore and aft line bench mark. A running fit shall be obtained between all studs of the gimbal ring and bracket mounting shown on Drawing 815-1197068 and the bushings of the same gimbal ring and bracket mounting and of the indicator into which the studs fit.

3.12.3 Types E and R, 60 and 400 Hz, double dial, concentric steering indicators. The types E and R indicators shall have one single speed dial and one 6-speed dial, both dials being in the same gear train. The servo mechanism shall be similar to that used on the bearing indicator (see 3.12.2). The arrangement for mounting shall be as specified in 3.12.1.

3.12.3.1 Dial. The single speed dial shall be the inner dial and shall be of 5-3/4 inch diameter. The six speed dial shall be the outer dial and shall be driven from the 36-speed synchro drive gears through a 6:1 gear reduction. The outer dial shall make six revolutions to one revolution of the inner dial. The outer dial, except for five divisions of the dial at the index, shall be covered by an opaque mask, to prevent distraction of the eye from the inner dial. The outer dial shall be graduated from 0 to 10 degrees over each one-sixth of its periphery with every 1/2 degree marked and every degree mark numbered from 0 to 9 in a continuing sequence, completely around the dial. The outer diameter of the outer dial shall be 6-7/8 inches. Between the inner and outer dials a white marked red illuminated diamond shaped index shall be provided for reading the ship's true heading. The dials shall have black background with white markings. The dials shall otherwise be in accordance with Drawing 9000-S6504-73687.

3.12.4 Types F and S, 60 and 400 Hz, double dial, concentric steering indicator (console mounted). The types F and S indicators shall be in accordance with the requirements for types E and R, except as follows:

3.12.4.1 The enclosure shall be nonwatertight and a mounting flange shall be provided for panel mounting of the enclosure. Four 13/32-inch diameter holes arranged symmetrically on either side of the centerline of the dial, when the dial is alined to 0 degrees, with their centers 7-7/16 inches apart, shall be provided in the flange for mounting.

3.12.4.2 No bezel shall be provided integral with the indicator; protection for the dial and servo system shall be provided by the steering console. A window assembly with the necessary gasket to preserve the splashproof integrity of the console shall be provided for mounting on the console cover and shall give an unimpaired view of the dials. When specified (see 6.2.1), sinuous course clock adapter rings shall be provided for mounting a sinuous

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course clock on the console. The adapter ring shall be mounted using the same holes provided in the console cover as for the window assembly and the same gasket as for the window assembly.

3.12.4.3 Access to the ship's course indicator will be from the top when the top of the steering console is raised.

3.12.4.4 No dimmer rheostat as specified in 3.7 shall be provided. The illumination control is provided in the console.

3.12.5 Type G, 60 Hz, double dial, course-to-steer indicator. The type G indicator shall have mounting features as specified in 3.12.1 and shall have dials with a black background and white markings.

3.12.5.1 The indicator shall have two concentric dials. The inner dial shall be positioned by a 60 Hz synchro motor and the outer dial by a servo unit consisting of a standard Navy synchro control transformer, a servo amplifier, and a servo motor. The inner dial is for own ship's heading indications while the outer dial is for course-to-steer indications.

3.12.5.2 The inner dial shall be in accordance with Drawing 9000-S6504-73687 except the major graduations shall be 1/4 inch, the intermediate graduations 1/8 inch, the minor graduations 1/16 inch and the numerals 3/16 inch. The inner dial shall have a minimum diameter of 5-1/2 inches.

3.12.5.3 The outer dial shall have no graduations engraved other than configuration of a diamond with a nominal 1/4-inch height engraved at the zero point.

3.12.5.4 A white marked, red illuminated fixed index like that specified in 3.12.3.1 shall be provided outside the outer dial at the zero point and shall be extended as a red illuminated lubber's line over the outer and inner dial markings.

3.12.6 Type H, 60 Hz, double dial, course-to-steer indicator (console mounted). The type H indicator shall be in accordance with the requirements for type G, except as specified herein.

3.12.6.1 The enclosure shall be nonwatertight and a mounting flange shall be provided for panel mounting of the enclosure. Four 13/32 inch diameter holes arranged symmetrically on either side of the centerline of the dial, when the dial is alined to 0 degrees, with their center 7-7/16 inches apart shall be provided in the flange for mounting.

3.12.6.2 No bezel shall be provided; protection for the dial and servo system will be provided by the steering console. No window assembly shall be provided.

3.12.6.3 Access to the ship's course indicator will be from the top when the top of the steering console is raised.

3.12.6.4 No dimmer rheostat as specified in 3.7 shall be provided. The illumination control is provided in the console.

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3.12.7 Type K, 60 Hz, single dial, bulkhead mounted indicator. The type K indicator shall have a single dial of 6-inch diameter driven by a 60 Hz synchro motor. The dial and index shall be identical to those described in 3.12.1.1. The dial shall have black background with white markings; the index shall be white, red illuminated and fixed. The mounting arrangement shall be as specified in 3.12.1.

3.12.8 Type L, miniature, 60 or 400 Hz, single dial indicator (universally mounted). The type L indicator shall be capable of universal mounting. The overall size shall not exceed the dimensions of a 6-inch square in the plan view of the dial or a depth of 6 inches.

3.12.8.1 Dial. The dial shall be 4 inches in diameter and shall be positioned by a synchro of a size to meet the accuracy requirements (see Drawing 9000-S6504-73687 for detailed information on the dial markings). A fixed index like that of 3.12.1.1 shall be provided in lieu of a fixed outer ring dial. The dial shall have a black background with white markings.

3.12.8.2 A dimmer rheostat shall be provided for the self-contained red internal illumination control but the design shall permit remote illumination control.

3.12.8.3 The frequency, 60 or 400 Hz, on which the ship's course indicator shall operate, shall be as specified (see 6.2.1).

3.12.9 Types M and T, 60 and 400 hz, single dial, bulkhead mounted indicators. The types M and T indicators shall be designed to mount directly on a bulkhead with the dial in a vertical position.

3.12.9.1 Dial. The inner dial shall be positioned by the synchro mechanism. The diameter of the inner dial and the effective outer diameter of its degree mark circle shall be 11.0 inches. Graduations shall be applied for every 5-degree and 10-degree increments, every 5-degree mark a line 1/4-inch in height, every 10-degree mark a line 1/2-inch in height. The width of the markings shall be 1/16 - 3/22 inch. Numerals of 1-inch height shall be placed inside the degree mark circle at intervals of 30 degrees commencing with 0 degrees. The height to width ratio of the numerals shall be in accordance with Drawing 9000-S6504-73687. The stroke width of the numerals shall be 1/16 inch. The numerals to cover 0-360 degrees shall read from 0-33 in increments of 3. In lieu of a fixed outer dial, a fixed red illuminated index such as that of 3.12.1.1, but having a nominal 1/2 inch base and a nominal 3/4 inch height, shall be used. The dial shall have a black background and white markings.

3.12.10 Type N, 60 or 400 Hz, as specified (see 6.2.1), single dial bearing, portable. The type N indicator shall be so constructed that when a Navy azimuth circle or bearing circle is attached, the cross hair of the azimuth circle or bearing circle shall be plainly visible against the background of the fixed and moving dials in any position in daylight, twilight or darkness. From the extreme top surface of the bezel ring to the faces of the moving and fixed dials described for this indicator, the distance shall be held between 3/4 and 1-1/4 inches.

3.12.10.1 Dial. The inner dial shall be in accordance with Drawing 9000-S6504-73687 as to markings and shall be positioned by a single size 23 or

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31 torque receiver synchro. Around the moving dial shall be a fixed ring dial graduated from 0 degree to 359 degrees in one-degree divisions with markings corresponding to those for the inner dial but with the numerals to the outside edge. This fixed ring dial shall be used in determining the bearing of an object relative to the ship. The dials shall have white background with black graduations to facilitate the taking of bearings.

3.12.10.2 The indicator shall have a rim of 70 - 30 percent copper-nickel alloy. The rim shall have a diameter of 7.800 inches $\pm$0.000/-0.005 inch, to permit use of a Navy azimuth circle or bearing circle.

3.12.10.3 Bearing indicator case and mounting. The case shall be so that the indicator will mount in the gimbal ring and bracket for submarines or in the aluminum gimbal ring and aluminum bracket of Drawing 815-1197068 for surface ships. The type of gimbal ring and bracket mounting to be supplied shall be specified "for submarines" or "for surface ships" (see 6.2.1). The gimbal ring and bracket mounting shall permit free motion of the type N indicator during a roll of 45 degrees or a pitch of 15 degrees from the vertical or during any combination of the two motions, and there shall be no obstruction in the line of sight when taking bearings under conditions of 20 degrees roll and 5 degrees pitch. A pedestal identical to the pedestal for the types B and P indicators shall be furnished when specified (see 6.2.1) for support of the bracket shown on Drawing 815-1197068.

3.12.10.3.1 The indicator shall be pendulous when mounted in the gimbal ring with a 1-1/2 pound weight mounted to simulate a Navy azimuth circle or bearing circle. The portion of the case beneath the top bezel and cover assembly shall be designed with a handle to permit the indicator to be conveniently carried, so the dial may be easily read while horizontal.

3.12.10.3.2 For ship's cable entry into the enclosure a stuffing tube shall be used which will accommodate type MHOF-7 cable of MIL-C-915. When a type N indicator is specified as being acquired for submarine service (see 6.2.1), a 5-foot long type MHOF-7 cable in accordance with MIL-C-915 shall be supplied by the contractor and shall be fitted through a stuffing tube, also supplied by the contractor, to each indicator. The free end of the cable shall be fitted with a connector plug, Military Part Number M26482/4-012 of MIL-C-26482 and MIL-C-26482/4. On each plug, the contractor shall supply protective covers of the same copper-nickel material as the plug. Each cover shall have an attaching chain.

3.12.10.4 Alinement. Dowels or alinement marks shall be used to allow disassembly of the indicator and its reassembly without allowing more than plus or minus 0.0025 inch shift in the relative bearing dial index or lubber's line relative to the ship's reference fore and aft line bench mark. Similarly, dowels or alinement marks shall be used to prevent more than plus or minus 0.0025 inch shift between the line defined by the fore and aft marks on the bezel and the ship's reference fore and aft line bench mark. A running fit shall be obtained between all the studs of the gimbal ring and bracket as shown on Drawing 815-1197068 and the bushing of the same gimbal ring and bracket and the bushings of the ship's course indicator into which the studs fit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the tests specified in table I.

4.2.1 For qualification of types B, N and P indicators, it will be necessary that the associated gimbal ring, bracket mounting and pedestal also be tested as required by note 1 of table I.

TABLE I. Qualification inspection.

Tests	Requirement reference	Test reference
General examination	MIL-E-16400	4.4
Accuracy	3.5	4.5.1 and 4.5.2
Synchronization	3.8	4.5.8
Oscillation	3.9	4.5.9
Radio frequency noise interference	MIL-E-16400, MIL-STD-461 class IIA, MIL-STD-462, and MIL-STD-463	MIL-E-16400 1/ MIL-STD-461 class IIA, MIL-STD-462 and MIL-STD-463
Permeability	3.10	4.5.10
Transient	3.4.1	MIL-E-16400
Airborne noise (all indicators shall meet grade C equipment limits of MIL-STD-740 in sound power levels for octave bands indicated, and MIL-E-16400 limits for other than low noise equipments)	MIL-E-16400 and MIL-STD-740	MIL-E-16400
Structureborne noise (type 3 equipment limits of MIL-STD-740 for low noise equipments)	MIL-E-16400 and MIL-STD-740	MIL-E-16400 2/
Enclosure (not applicable to types F, H and S)	MIL-E-16400	MIL-E-16400
Salt spray (not applicable to types F, H and S)	MIL-E-16400	MIL-E-16400

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TABLE I. Qualification inspection. - Continued

Tests	Requirement reference	Test reference
Accelerated life test Includes vibration test for vital equipment 2/ (4.5.3), power supply tolerance (4.5.4), temperature and humidity (4.5.5), inclination (4.5.6), shock 2/ (4.5.7), dielectric strength (for all but rotating components) (MIL-E-16400), insulation resistance (for all but rotating components) (MIL-E-16400).	MIL-E-16400	MIL-E-16400

1/ Operating condition for the test shall be a constant velocity of 2 degrees per second.

2/ The gimbal ring, bracket mounting and pedestal shall be tested with the associated bearing indicator or a dummy weight installed for these tests. They shall perform their principal functions and shall suffer no fracture, permanent deformation, or amplification of vibration at any frequency or amplitude required herein, such that observation, operation, or adjustment of an indicator mounted therein would be impossible or excessively difficult during the test.

4.2.2 Samples for qualification. One indicator of each type shall be subjected to qualification testing either individually or representatively as indicated below:

Individually	Representatively
Type A, B, E, F, G, H K, L, M, N, O, P, R, S, and T	Type B - Will include qualification of types A, E, F, K, M and N (see 4.2.1) with contingency that the first production types K and M indicators and the first production type N indicator with its associated gimbal ring and bracket shall be required to pass all group C tests specified in table III. The test results shall be forwarded to NAVSEA for approval. Type G - Will include qualification of type H and vice versa. Type P - Will include qualification of types O, R, S and T with the contingency that the first production type T indicator shall be required to pass all group C tests specified in table III, unless the design of the type T indicator is like the design of type M indicator and the type M indicator has passed all group C tests.

4.3 Quality conformance inspection.

4.3.1 Group A. Each indicator, gimbal ring, bracket mounting and pedestal on the contract or order shall be subjected to the group A examination and tests specified in table II.

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over the range 40 to 130° F, the accuracy shall be as specified in 3.13.3.1 and 3.13.3.2;

TABLE II. Group A examination and tests.

Examination and tests	Requirement reference	Test reference
General examination 1/	MIL-E-16400	4.4
Dielectric strength 2/	MIL-E-16400	MIL-E-16400
Insulation resistance 2/	MIL-E-16400	MIL-E-16400
Accuracy	3.5	4.5.1 and 4.5.2
Synchronization	3.8	4.5.8
Oscillation	3.9	4.5.9

1/ To be applied also to the gimbal ring and bracket mounting and to the pedestal.
2/ Not to be applied to rotating components.

4.3.2 Group C. One complete indicator of or representative of each type and one gimbal ring and bracket mounting or pedestal shall be selected and subjected to the group C tests specified in table III only when the basic design of the equipment or the material of a vital part has been changed from that of the equipment qualified.

TABLE III. Group C tests.

Tests	Requirement reference	Test reference
Radio frequency noise interference	MIL-E-16400, MIL-STD-461 class IIA, MIL-STD-462, and MIL-STD-463	MIL-E-16400 1/ MIL-STD-461 class IIA, and MIL-STD-463
Permeability	3.10	4.5.10
Transient	3.4.1	MIL-E-16400
Enclosure (not applicable to types F, H and S)	MIL-E-16400	MIL-E-16400
Salt spray (not applicable to types F, H and S)	MIL-E-16400	MIL-E-16400
Airborne noise (all indicators shall meet grade C equipment noise limits of MIL-STD-740 in sound power levels for octave bands indicated, and MIL-E-16400 limits for other than low noise equipments)	MIL-E-16400 and MIL-STD-740	MIL-E-16400 1/
Structureborne noise (type 3 equipment noise limits of MIL-STD-740 for low noise equipment)	MIL-E-16400 and MIL-STD-740	MIL-I-16400 2/

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TABLE III. Group C tests. - Continued

Tests	Requirement reference	Test reference
Accelerated life test Includes vibration test for vital equipment 2/ (4.5.3), power supply tolerance (4.5.4) temperature and humidity (4.5.5), inclination (4.5.6), shock 2/ (4.5.7), dielectric strength (for all but rotating components) (MIL-E-16400), insulation resistance (for all but rotating components) (MIL-E-16400).	MIL-STD-740	MIL-E-16400

1/ Operating condition for the test shall be a constant velocity of 2 degrees per second.

2/ To be applied also to the gimbal ring, bracket mounting and pedestal. The gimbal ring, bracket mounting and pedestal shall be tested with the associated bearing indicator or a dummy weight installed for these tests. They shall perform their principal functions and shall suffer no fracture, permanent deformation, or vibration at any frequency or amplitude required herein, such that observation, operation, or adjustment of an indicator mounted therein would be impossible or excessively difficult during the test.

4.3.3 Rejection. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A or C test shall be counted as a defect, and the equipment shall not be offered for delivery.

4.4 General examination. The general examination shall be conducted in accordance with MIL-E-16400. No operating test shall be required.

4.5 Tests.

4.5.1 Static accuracy. The equipment under test shall be operated with normal input signals. The static accuracy of the equipment shall be determined by turning the dummy director in one direction 360 degrees in 6 steps and measuring the error at each step according to the method specified for the dynamic accuracy test (see 4.5.2). The same procedure shall be repeated in the opposite direction of rotation. The maximum error as indicated by the error voltage shall not exceed the values specified in 3.5.

4.5.2 Dynamic accuracy.

4.5.2.1 Servo type. The accuracy of the ship's course indicators shall be determined by operating the equipment with a one speed or a one and 36-speed input, as applicable, from a dummy director or equivalent synchro device. The rotor leads of the high speed synchro control transformer (one speed for single speed indicators) of the equipment under test shall be connected to a cathode ray oscillograph or other equivalent accurate voltage indicating instrument. The voltage measured across the rotor leads of the synchro control transformer represents the follow-up error of the equipment under test. When there is no voltage across the rotor leads of the synchro control transformer, there is no

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displacement of the control transformer, with respect to the synchro transmitter in the dummy director; therefore no follow-up error exists in the ship's course indicator under this condition. Likewise, a voltage indication across the rotor leads of the synchro control transformer denotes a displacement of the control transformer with respect to the transmitter in the dummy director, therefore an error proportionate to the magnitude of the measured voltage exists in the ship's course indicator. In order to determine the error in terms of degrees and minutes for the purpose of this test, the angular deviations of the rotor of the control transformer with respect to the rotor of the synchro transmitter shall be calibrated in terms of voltage values as determined from the oscilloscope, oscillograph or equivalent device.

4.5.2.2 Synchro type. The dynamic accuracy shall be measured by means of the reflected load on the synchro bus. A type 5CT synchro control transformer shall be directly geared to a synchro transmitter in the dummy director. The stator leads of the 5CT synchro control transformer shall be connected to the stator leads of the synchro transmitter in the dummy director, in parallel with the synchro in the indicator. The voltage across the rotor leads of the 5CT control transformer (excluding voltage due to synchro electrical error) denotes a displacement of the synchro in the equipment from the synchro transmitter in the dummy director; therefore an error proportionate to the magnitude of the measured voltage exists in the ship's course indicator. In order to determine the error in terms of degrees and minutes, for the purpose of this test, the angular deviations of the rotor of the synchro, with respect to the rotor of the synchro transmitter, shall be calibrated in terms of voltage values as determined from the oscilloscope, oscillograph, or equivalent device.

4.5.2.3 The dynamic accuracy shall be determined by operating the dummy director under the following conditions (all speeds are of the 1-speed synchro):

(a) Constant velocity 1 degree per second
(b) Constant velocity 6 degrees per second

} Perform in CW direction, then in CCW direction; or vice versa.

(c) Yaw 4.0-degree amplitude 4.5 second period.

The dynamic error as indicated by the voltage across the rotor leads of the synchro control transformer shall not exceed the values specified in 3.5.

4.5.3 Vibration test. The vibration test shall be conducted in accordance with MIL-E-16400. The equipment shall be operated with a constantly varying input signal of simple harmonic motion of 4.0 degree amplitude and 4.5 second period. During the test the equipment shall meet the dynamic accuracy (simple harmonic motion) test as specified in 4.5.2. After the test, the equipment shall meet the static and dynamic accuracy tests as specified in 4.5.1 and 4.5.2.

4.5.4 Power supply tolerances. The equipment shall undergo and meet the static and dynamic accuracy tests as specified in 4.5.1 and 4.5.2 while being supplied power at the upper and lower limits specified in MIL-E-16400.

4.5.5 Temperature and humidity. The equipment shall be operated the same as for the vibration test as specified in 4.5.3. A type B, N, and P indicator shall be tested at a low temperature of minus 28°C, all other indicators shall

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be tested at a low temperature of 4.4°C. Types of indicators shall be tested uninterruptedly for the high temperature portion of the accelerated life test of 60°C. The type B, N, and P indicator shall, in addition to the above test, be tested for compliance with the requirements as specified in MIL-E-16400 for extreme temperature for exposed locations. Any time during the temperature and humidity test the equipment shall meet the dynamic accuracy (simple harmonic motion) test as specified in 4.5.2. After the test, the equipment shall meet the static and dynamic accuracy tests as specified in 4.5.1 and 4.5.2.

4.5.5.1 Extended storage. The equipment shall be tested in accordance with MIL-E-16400. After stabilizing from exposures to the extreme temperature limits the equipment shall meet the static and dynamic accuracy tests as specified in 4.5.1 and 4.5.2.

4.5.6 Inclination test. The equipment shall be operated the same as for the vibration test specified in 4.5.3 and shall, during the test, meet the dynamic accuracy (simple harmonic motion) test as specified in 4.5.2.

4.5.7 Shock test. The equipment shall be operated the same as for the vibration test specified in 4.5.3. The equipment shall after each shock meet the dynamic accuracy (simple harmonic motion) test as specified in 4.5.2. After the complete shock test, the equipment shall meet the static and dynamic accuracy tests as specified in 4.5.1 and 4.5.2.

4.5.8 Synchronization. Tests shall be made to insure that the equipment will be self-synchronous as specified in 3.8.

4.5.9 Oscillations. Tests shall be made to insure that oscillations are controlled as specified in 3.9.

4.5.10 Permeability. Material shall be checked, after fabrication, using a permeability indicator, low-mu (Go-No-Go) in accordance with MIL-I-17214 (see 3.10).

4.6 Inspection of packaging. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Domestic shipment and early equipment installation (see 5.3.1).

5.1.1 Ship's course indicators.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the contractor's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

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5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. The onboard repair parts shall be preserved and packaged by level A; packed and marked by level C in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the contracting activity (see 6.2.1).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when acquisition is made.

5.2.1.1 Preservation and packaging, packing and marking. The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified and marked in accordance with MIL-E-17555 (see 5.3.2).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the contracting activity (see 6.2.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. Unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc., with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

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6. NOTES

6.1 Intended use. These ship's course indicators are used to indicate the master gyrocompass headings in various locations throughout Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required and if type L or N is required, whether 60 or 400 Hz is required (see 1.2, 3.12.8.3 and 3.12.10).
(c) Quantity of items required.
(d) Type of mounting for types B and P bearing indicators (see 3.12.2.4).
(e) Whether sinuous course clock adapter rings should be furnished (see 3.12.4.2).
(f) Type of mounting for type N indicator; that is, gimbal ring and bracket "for submarines" or "for surface ships" and pedestal if required (see 3.12.10.3).
(g) When a type N indicator is required for "submarines" (see 3.12.10.3.2).
(h) Equipments required to meet structureborne noise limits of MIL-STD-740 for type 3 equipment.
(i) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).
(j) When polystyrene (loose-fill) material is approved (see 5.3.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-15676 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Provisioning. Provisioning technical documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions

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as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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